LECTURE NOTES MATH255

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1. Lecture 17

In this class we introduce the notion of a Euclidean domain, and prove that they are always PIDs.

1.1. Euclidean norms and Euclidean domains.

A Euclidean domain R is an integral domain that is equipped with a *Euclidean norm*. So before we can define a Euclidean domain, we must first introduce the notion of a Euclidean norm.

Definition 1.1. *Let R be an integral domain. A Euclidean norm on R is a function $\nu : R \setminus \{0\} \to \mathbb{N}_0 = \{0, 1, 2, ...\}$ that satisfies the two properties:*

(i) If $a, b \in R$, $b \neq 0$, then there exists $q, r \in R$ such that $a = bq + r$ and either $r = 0$ or $\nu(r) < \nu(b)$.
(ii) If $a, b \in R \setminus \{0\}$ then $\nu(ab) \geq \nu(a)$.

Remark 1. The first part of the definition is essentially telling us that if we use ν to measure 'size' of elements in $R \setminus \{0\}$ then R has some notion of a division algorithm, where if we divide a by b then we obtain a remainder r where r is 'smaller' than b (i.e., either $r = 0$ or $\nu(r) < \nu(b)$).

Example 1.1. The integers $\mathbb{Z}$ have a Euclidean norm $\nu(a) = |a|$ where $|a|$ is the absolute value of $a \in \mathbb{Z}$. Indeed, by the standard division algorithm we can write any $a \in \mathbb{Z}$ as $a = bq + r$ where $r \in \{0, ..., |b| - 1\}$ which implies that $\nu(a) = |a|$ satisfies the first property of being a Euclidean norm. To see the second property, note that if $m, n \in \mathbb{Z} \setminus \{0\}$ then $|mn| = |n| \cdot |m| \geq |n|$ since $|m| \geq 1$.

Example 1.2. Let F be any field, and consider the polynomial ring $F[x]$. For any non-zero polynomial $f(x) \in F[x]$ we define $\nu(f(x)) = \deg(f(x))$. By the division algorithm for polynomials it follows that ν satisfies the first property of being a Euclidean norm. The second property follows by noting that if $f(x), g(x) \in F[x]$ are non-zero polynomials then $\deg(f(x)g(x)) = \deg(f(x)) + \deg(g(x))$ so $\nu(f(x)g(x)) = \nu(f(x)) + \nu(g(x)) \geq \nu(f(x))$.

Definition 1.2. *Let R be an integral domain. We say that R is a Euclidean domain if there is some Euclidean norm ν on R.*

Theorem 1.1. *Any Euclidean domain is a PID.*

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Proof. Let R be a Euclidean domain with Euclidean norm ν. Let $I \subset R$ be an ideal. We will prove that I is principal. Consider first the case when $I = \{0\}$, then $I = \langle 0 \rangle$ so I is principal. We now assume that $I \neq \{0\}$, so there exists some non-zero element in I. We define $a \in I \setminus \{0\}$ as any element that satisfies:

$$\nu(a) = \min_{x \in I \setminus \{0\}} \nu(x), \tag{1.1}$$

note that such an element exists since $\mathrm{Im}(\nu) \subset \mathbb{N}_0$ is some subset that is bounded from below (by 0). Let $b \in I$ be arbitrary. Using the first property of a Euclidean norm we can write:

$$b = aq + r \tag{1.2}$$

where either $r = 0$ or $\nu(r) < \nu(a)$. Notice that $r = b - aq \in I$ since $a \in I$, $b \in I$, so we can not have $r \neq 0$ and $\nu(r) < \nu(a)$ since this would contradict minimality of a. It follows that $r = 0$ and we obtain $b = aq \in \langle a \rangle$. That is, we have shown that if $b \in I$ is arbitrary, then $b \in \langle a \rangle$ so $I \subset \langle a \rangle$. Conversely, note that $a \in I$ so $\langle a \rangle \subset I$. Combined we obtain $I = \langle a \rangle$, and since I was an arbitrary (non-zero) ideal we deduce that every ideal I is principal. ■

Theorem 1.2. *Let R be a Euclidean domain with Euclidean norm ν. Then $\nu(1)$ is the minimal value of ν, and an element $a \in R \setminus \{0\}$ is a unit if and only if $\nu(a) = \nu(1)$.*

Proof. Let $a \in R \setminus \{0\}$, then $\nu(a) = \nu(1 \cdot a) \geq \nu(1)$ which proves that $\nu(1)$ is the minimal value of ν. Assume now that $\nu(u) = \nu(1)$. Using the first property of the Euclidean norm with $a = 1$ and $b = u$ we obtain $1 = qu + r$ where either $r = 0$ or $\nu(r) < \nu(u)$. However, since $\nu(u) = \nu(1)$ is the minimal value of ν we can not have $\nu(r) < \nu(u)$, so $r = 0$. That is, we have $1 = qu$ which implies that u is a unit (with multiplicative inverse q). ■

We will give one more example of a Euclidean domain R. Let i be the imaginary unit, defined by $i^2 = -1$, and define the *Gaussian integers* by:

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}. \tag{1.3}$$

We leave it as an exercise to check that $\mathbb{Z}[i]$ is an integral domain with addition and multiplication from $\mathbb{C}$. On $\mathbb{Z}[i]$ we define the map $\nu : \mathbb{Z}[i] \to \mathbb{N}_0$ by $\nu(a + bi) = a^2 + b^2 = |a + bi|^2$ (note that ν is also defined at 0, this will be convenient for now, but at the end we will restrict ν to $\mathbb{Z}[i] \setminus \{0\}$ to be consisitent with the definition of a Euclidean norm).

Lemma 1.1. *The function ν satisfies $\nu((a + bi)(c + di)) = \nu(a + bi)\nu(c + di)$, and $\nu(a + ib) = 0$ if and only if $a + ib = 0$.*

Remark 2. This lemma actually shows that ν is a multiplicative norm, which will be studied in the next class.

Proof. We recall that if $z, w \in \mathbb{C}$ then $|zw| = |z| \cdot |w|$. It follows that if $z, w \in \mathbb{Z}[i]$ then:

$$\nu(zw) = |zw|^2 = |z|^2 |w|^2 = \nu(z)\nu(w) \tag{1.4}$$

which proves the first part of the lemma. Note that $|z|^2 = 0$ if and only if $z = 0$, this immediately implies the last claim of the lemma. ∎

Theorem 1.3. *The map* $\nu : \mathbb{Z}[i]\backslash\{0\} \to \mathbb{N}$ *is a Euclidean norm, so* $\mathbb{Z}[i]$ *is a Euclidean domain.*

Proof. Note that if $z, w \in \mathbb{Z}[i] \setminus \{0\}$ then $\nu(z), \nu(w) \geq 1$ from the previous lemma (since $\nu(z), \nu(w)$ are non-zero non-negative integers). Using the multiplicative property it follows:

$$\nu(zw) = \nu(z)\nu(w) \geq \nu(z) \cdot 1 = \nu(z) \tag{1.5}$$

which proves the second property of being a Euclidean norm.

It remains to show that if $z, w \in \mathbb{Z}[i]$, $w \neq 0$, then we find $q, r \in \mathbb{Z}[i]$ such that $z = qw + r$ and either $r = 0$ or $\nu(r) < \nu(w)$. Let $\sigma + i\tau = z/w$ and let $n, m \in \mathbb{Z}[i]$ be such that $|\sigma - n| \leq 1/2$ and $|\tau - m| \leq 1/2$. We define $q = n + im$ and calculate:

$$|z - qw|^2 = \left| z - \left(\frac{z}{w} + \left(q - \frac{z}{w} \right) \right) w \right|^2 = \left| \left(q - \frac{z}{w} \right) w \right|^2 =$$
$$|w|^2 \left((\sigma - n)^2 + (\tau - m)^2 \right) \leq |w|^2(1/4 + 1/4) = |w|^2/2.$$

It follows that if we define $r = z - qw$ then $\nu(r) \leq \nu(w)/2 < \nu(w)$ (where the last inequality use that $\nu(w) \neq 0$ since $w \neq 0$), and $x = qw + r$. ∎

Corollary 1.3.1. *The Gaussian integers form a PID (and a UFD), and the units in* $\mathbb{Z}[i]$ *are precisely* $1, -1, i, -i$.

Remark 3. Since $\mathbb{Z}[i]$ is a UFD we get a notion of *Gaussian primes*, which are the irreducible elements in $\mathbb{Z}[i]$. These Gaussian primes are then similar to the normal primes in the sense that any Gaussian integer can be written as a product of Gaussian primes uniquely (up to multiplying with a unit $1, -1, i, -i$).

Proof. Since $\mathbb{Z}[i]$ is a Euclidean domain it is a PID, and every PID is a UFD, so $\mathbb{Z}[i]$ is also a UFD. To calculate the units in $\mathbb{Z}[i]$ we need to find which $a + ib \in \mathbb{Z}[i]$ satisfy $|a + ib|^2 = a^2 + b^2 = 1$. However, since a, b integers we can only solve this equation if $a = \pm 1$ and $b = 0$ or $a = 0$ and $b = \pm 1$. This corresponds to $1, -1, i, -i \in \mathbb{Z}[i]$. ∎

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